Wave Problems

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Motivation and Context

- Some systems are traditionally described by equations (PDE)
- Remember PCSE, where the crop growth was simulated based on PDE systems
- For most cases, we have data that covers part of the phenomenon
- For multiple reasons, the equations often cannot fully describe the data
- Two problems are usually tackled on those systems: Forward and Inverse

Motivation and Context

- Forward problems: in the case of dynamical systems, take the form of (uni- or multivariate) forecasting
- Inverse problems: retrieve information about the system, using the observed data
- The definition of those problems is general enough to usually be tackled by two sides:
  - Mathematical/physical modelization
  - Machine Learning

Specifically: waves

- My focus has been set on two wave equations:
  - Acoustic: $\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$
  - Elastic: $\rho \frac{\partial^2 u}{\partial t^2} = (\lambda + \mu) \nabla \Delta u + \mu \nabla^2 u$
- The main difference lies in the fact that we consider the propagation field to be deformable or not.
- In both cases, boundary conditions over u and its first two derivatives have to be defined.

Acoustic Wave

- $\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$
- The form is fairly simple
- The only parameter is c, the propagation speed field
- Impossible to fully and continuously describe in reality
- Solving it through time requires discretization, in space, and in time
- Multiple solvers exist (finite differences, finite volumes, finite elements, pseudo-spectral, spectral)

Elastic Wave

- $\rho \frac{\partial^2 u}{\partial t^2} = (\lambda + \mu)\nabla \Delta u + \mu \nabla^2 u$
- The form is much more complex (the gradient of the divergence appears)
- There are three parameters, λ and μ, which are called the *Lamé Parameters*, and which describe the deformability of the propagation field, and ρ, which describes the density of the field
- Same solving problems as for acoustic wave
- Elastic wave can be mathematically decomposed into two waves, whose deformation direction is orthogonal: the Primary and Secondary waves, which have different propagation speeds: they depend on the Lamé Parameters

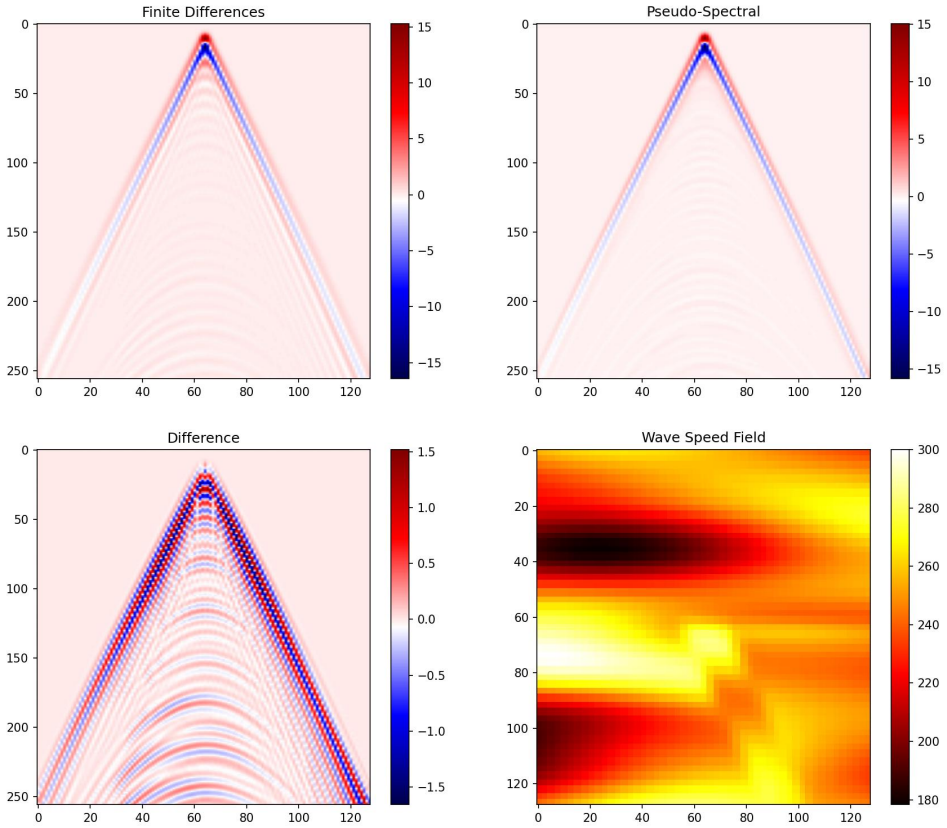
Data collection

- Two different measurement devices are currently used:
  - Microphones/Seismometers
  - Optical fiber cables
- There is no complete spatial coverage
- There is no way to fully characterize the parameters c, λ and μ
- It can be partially done by *inversion*

What the data looks like

- For unique seismometers, the data is usually uni-, bi- (tri-) variate time series, showing the ground displacement, where the seismometer is put, in two (three) spatial directions.
- Multiple seismometers can be concatenated to get more than trivariate series, allowing more spatial coverage
- For optical fiber cables, the data is similar to multiple seismometers put aside
- The data of multiple seismometers/optical fiber is often displayed as a *trace*

Trace



Trace examples and propagation field example, as would be obtained by a line of seismometers or by an optical fiber cable, in 2D

Problems

- Forward:
  - Given a description of the parameters, forecast the propagation of a wave
  - Given early measurements, forecast the evolution
  - Given simulated measurements, enhance the simulation to get closer to observed measurements
- Inverse
  - Given observed measurements, retrieve the parameters of the equations
  - Given observed measurements, determine a model that outperforms the traditional approaches

Link with DT

- On the modelisation/simulation part of the DT framework
- This framework gives a complete playground to experiment with physics-based ML methods (topic for a MLG talk?):
  - Differentiable Programming
  - PINNs
  - Neural Operators
  - ...
- The framework is (hopefully) general enough to allow changing the studied system and replacing the PDEs